# Helix Forge Research Ablation Design Methods Study

HELIX FORGE LAB / p04-full-006131 / RUN R184 / 2026-07-13

| ALPHA F1 | PRECISION | RECALL | BETA F1 |
|---|---|---|---|
| 0.873 | 0.881 | 0.865 | 0.846 |

## Decision context

This report records a bounded ablation design evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

## Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

## Lineage

Upstream evidence: p04-full-000252 (notebooks/exports/2026-07-13/helix-forge-research/benchmark-design/specifications/helix-forge-research-benchmark-design-slices-v01.js

| Slice | Records | Precision | Recall | F1 |
|---|---|---|---|---|
| slice-01 | 168 | 0.870 | 0.873 | 0.871 |
| slice-02 | 248 | 0.887 | 0.849 | 0.868 |
| slice-03 | 212 | 0.869 | 0.862 | 0.865 |
| slice-04 | 165 | 0.886 | 0.875 | 0.880 |
| slice-05 | 160 | 0.868 | 0.851 | 0.859 |